

Algebra II with Trigonometry

Chapter 7.4 & 7.5

Study Guide (Gemini)

Saved tutoring response with MathJax rendering. Original PDF file:
[4626701_alg-2trig-h-chp-7-4-7-5-note-docx.pdf](#)

Here is a comprehensive study guide and breakdown of the provided document, created just for you!

1. Summary of the Document

This document is a guided worksheet/notes packet for an Algebra 2 Trigonometry/Honors class (Chapters 7.4 & 7.5). It covers three primary methods for solving trigonometric equations: 1. **Basic Equations:** Isolating the trigonometric function and finding general solutions by adding integer multiples of the period. 2. **Factoring:** Moving all terms to one side, factoring the expression, and using the Zero-Product Property. 3. **Using Identities & Multiple Angles:** Simplifying complex equations using trigonometric identities (like Pythagorean or angle addition formulas) or solving for a multiple angle first before dividing to find the base angle.

The document also provides four sets of example problems to practice these specific techniques.

2. Core Definitions, Theorems, and Formulas

- **Trigonometric Equation (Definition):** An equation that contains trigonometric functions.
- **Basic Trigonometric Equation (Definition):** An equation of the form $T(\theta) = c$, where T is a trigonometric function and c is a constant.
- **Zero-Product Property (Theorem):** If $AB = 0$, then $A = 0$ or $B = 0$.
- **Identities and Formulas Referenced:**
 - *Pythagorean Identities* (e.g., $\sin^2 \theta + \cos^2 \theta = 1$, $1 + \cot^2 \theta = \csc^2 \theta$)

- *Double-Angle or Half-Angle Formulas*
- *Addition or Subtraction Formulas* (e.g.,
 $\sin(A - B) = \sin A \cos B - \cos A \sin B$)
- *Sum-to-Product or Product-to-Sum Formulas*

3. Worked Study Guide

Let's walk through the three main strategies for solving trigonometric equations discussed in the text.

Strategy 1: Solving Basic Trigonometric Equations

Goal: Isolate the trig function to look like $T(\theta) = c$.

- **Step 1:** Find the solutions within one single period (usually $[0, 2\pi)$ for sine/cosine or $[0, \pi)$ for tangent).
- **Step 2:** Add integer multiples of the period to write the general solution (using n as an integer).

Example: Solve $\sin \theta = -\frac{\sqrt{3}}{2}$ and list six specific solutions. (From *Example 1a*)
1. Find solutions in one period: The sine function is negative in Quadrants III and IV. The base reference angle where $\sin(\theta) = \frac{\sqrt{3}}{2}$ is $\frac{\pi}{3}$.

- Quad III: $\pi + \frac{\pi}{3} = \frac{4\pi}{3}$
- Quad IV: $2\pi - \frac{\pi}{3} = \frac{5\pi}{3}$

2. Add integer multiples of the period (2π for sine):

- General Solutions: $\theta = \frac{4\pi}{3} + 2\pi n$ and $\theta = \frac{5\pi}{3} + 2\pi n$

3. List six specific solutions: Plug in integers for n (e.g., $n = 0, 1, -1$).

- If $n = 0$: $\frac{4\pi}{3}, \frac{5\pi}{3}$
- If $n = 1$: $\frac{10\pi}{3}, \frac{11\pi}{3}$
- If $n = -1$: $-\frac{2\pi}{3}, -\frac{\pi}{3}$

Strategy 2: Solving by Factoring

Goal: Group everything on one side of the equals sign so it equals zero, factor, and split using the Zero-Product Property.

Example: Solve $\sec \theta(2 \cos \theta - \sqrt{2}) = 0$ (From *Example 2c*) 1. **Apply Zero-Product Property:** Set each factor to zero.

- $\sec \theta = 0$ OR $2 \cos \theta - \sqrt{2} = 0$

2. Solve the resulting basic equations:

- $\sec \theta = 0$: The secant graph never crosses 0 (its range is $y \geq 1$ or $y \leq -1$). This part yields **no solution**.
- $2 \cos \theta = \sqrt{2} \implies \cos \theta = \frac{\sqrt{2}}{2}$: The cosine function is positive in Quadrants I and IV.
- General solutions: $\theta = \frac{\pi}{4} + 2\pi n$ and $\theta = \frac{7\pi}{4} + 2\pi n$.

Strategy 3: Using Identities and Multiple Angles

Goal: Use a known trig identity to simplify the equation into a single trig function, or solve for the "inside" of a multiple angle first.

Example: Find all solutions for $2 \sin(3\theta) + 1 = 0$. (From *Example 4a*) 1.

Isolate the trig function: $2 \sin(3\theta) = -1 \implies \sin(3\theta) = -\frac{1}{2}$ 2. **Solve for the MULTIPLE of the angle first:** Treat " 3θ " as its own variable.

Where does sine equal $-1/2$? In Quadrants III and IV.

- $3\theta = \frac{7\pi}{6} + 2\pi n$
- $3\theta = \frac{11\pi}{6} + 2\pi n$

3. Divide to solve for the angle (θ): Divide everything by 3.

- $\theta = \frac{7\pi}{18} + \frac{2\pi n}{3}$
- $\theta = \frac{11\pi}{18} + \frac{2\pi n}{3}$

4. Practice Problems

Here are five practice problems directly from the document.

Problem 1: Solve $3 \sin^2 \theta - 1 = 0$ (From Example 2a) **Answer:** Isolate $\sin^2 \theta$: $\sin^2 \theta = \frac{1}{3}$ Take the square root of both sides: $\sin \theta = \pm \frac{1}{\sqrt{3}} = \pm \frac{\sqrt{3}}{3}$
General Solution: $\theta = \arcsin(\frac{\sqrt{3}}{3}) + \pi n$ and $\theta = -\arcsin(\frac{\sqrt{3}}{3}) + \pi n$

Problem 2: Solve $3 \tan^3 \theta = \tan \theta$ (From Example 2g) **Answer:** Move terms to one side: $3 \tan^3 \theta - \tan \theta = 0$ Factor out $\tan \theta$: $\tan \theta(3 \tan^2 \theta - 1) = 0$
Zero-Product Property: 1) $\tan \theta = 0 \implies \theta = \pi n$ 2) $3 \tan^2 \theta - 1 = 0 \implies \tan^2 \theta = \frac{1}{3} \implies \tan \theta = \pm \frac{\sqrt{3}}{3}$ The solutions for tangent equaling $\pm \frac{\sqrt{3}}{3}$ are $\frac{\pi}{6}$ and $\frac{5\pi}{6}$ plus multiples of π . General Solution: $\theta = \pi n$, $\theta = \frac{\pi}{6} + \pi n$, and $\theta = \frac{5\pi}{6} + \pi n$.

Problem 3: Solve $\csc^2 \theta = \cot \theta + 3$ (From Example 2i) **Answer:** Use the Pythagorean Identity ($\csc^2 \theta = 1 + \cot^2 \theta$): $1 + \cot^2 \theta = \cot \theta + 3$ Move terms to one side to form a quadratic: $\cot^2 \theta - \cot \theta - 2 = 0$ Factor: $(\cot \theta - 2)(\cot \theta + 1) = 0$ Zero-Product Property: 1) $\cot \theta = 2 \implies \tan \theta = \frac{1}{2} \implies \theta = \arctan(\frac{1}{2}) + \pi n$ 2) $\cot \theta = -1 \implies \tan \theta = -1 \implies \theta = \frac{3\pi}{4} + \pi n$

Problem 4: Solve $\sin(2\theta) \cos \theta - \cos(2\theta) \sin \theta = \frac{\sqrt{3}}{2}$ (From Example 4f)

Answer: Use the Subtraction Formula for Sine [

$\sin(A - B) = \sin A \cos B - \cos A \sin B$]:

$\sin(2\theta - \theta) = \frac{\sqrt{3}}{2} \implies \sin \theta = \frac{\sqrt{3}}{2}$ General Solution: $\theta = \frac{\pi}{3} + 2\pi n$ and $\theta = \frac{2\pi}{3} + 2\pi n$

Problem 5: Solve $\sec(\frac{\theta}{2}) = \cos(\frac{\theta}{2})$ (From Example 4b) **Answer:** Rewrite secant using its reciprocal identity: $\frac{1}{\cos(\frac{\theta}{2})} = \cos(\frac{\theta}{2})$ Multiply both sides by $\cos(\frac{\theta}{2})$: $1 = \cos^2(\frac{\theta}{2})$ Take the square root: $\cos(\frac{\theta}{2}) = \pm 1$ Cosine equals ± 1 at multiples of π : $\frac{\theta}{2} = \pi n$ Multiply by 2 to solve for θ : $\theta = 2\pi n$

5. Assumptions and Ambiguities Resolved

- **Missing Text/OCR Errors:** The provided OCR misread several equations by dropping the trigonometric functions (e.g., reading "3 θ " instead of "3 $\sin^2 \theta$ "). I resolved this by directly reading the mathematics from the provided images rather than relying on the flawed OCR text.

- **Scope of Solutions:** In Examples 1 and 2, the instructions just say "Solve the given equation." It is standard practice in trigonometry that if an interval (like $[0, 2\pi)$) is not specified, you must provide the *general solution* covering all infinitely many angles (adding $+2\pi n$ or $+\pi n$). I assumed general solutions were required for everything unless noted otherwise.
- **Example 3 Requirement:** Example 3 specifies "Use a graphing device." Because a graphing calculator is required to find the intersections of $y = \sin x$ and $y = x^3$, I excluded this specific problem from the algebraic practice problems above to ensure you could do them purely by hand.